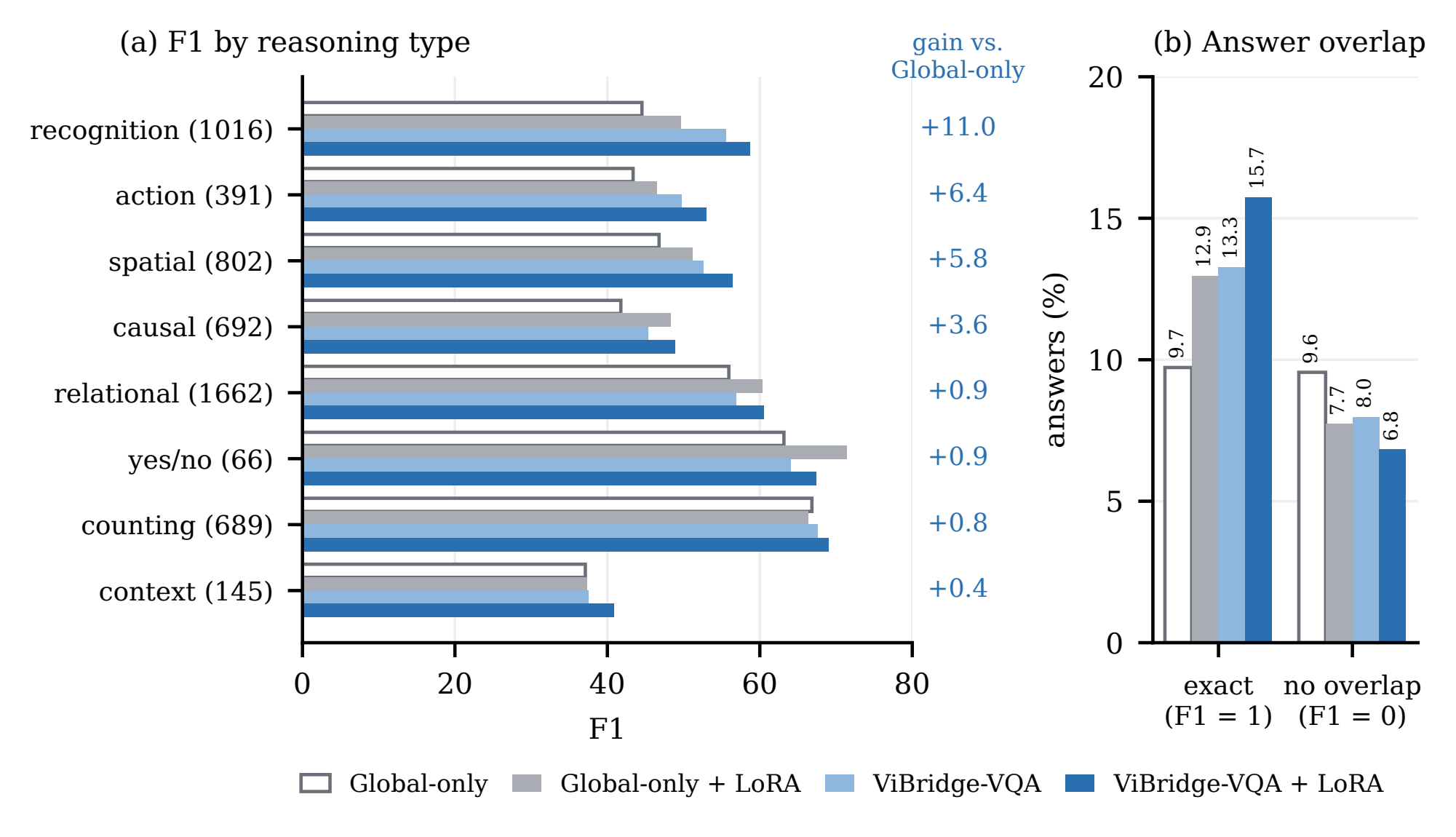
(a) F1 by reasoning type
gain vs. Global-only
recognition (1016)
action (391)
spatial (802)
causal (692)
relational (1662)
yes/no (66)
counting (689)
context (145)
+11.0
+6.4
+5.8
+3.6
+0.9
+0.9
+0.8
+0.4
0
20
40
60
80
F1
(b) Answer overlap
answers (%)
0
5
10
15
20
9.7
12.9
13.3
15.7
9.6
7.7
8.0
6.8
exact (F1 = 1)
no overlap (F1 = 0)
Global-only
Global-only + LoRA
ViBridge-VQA
ViBridge-VQA + LoRA